# Practice workbook · Identify urinary system and nephron structures

Naming, labelling, drawing or matching kidney, ureter, bladder, urethra and nephron parts, including assigning functions to structures in a table.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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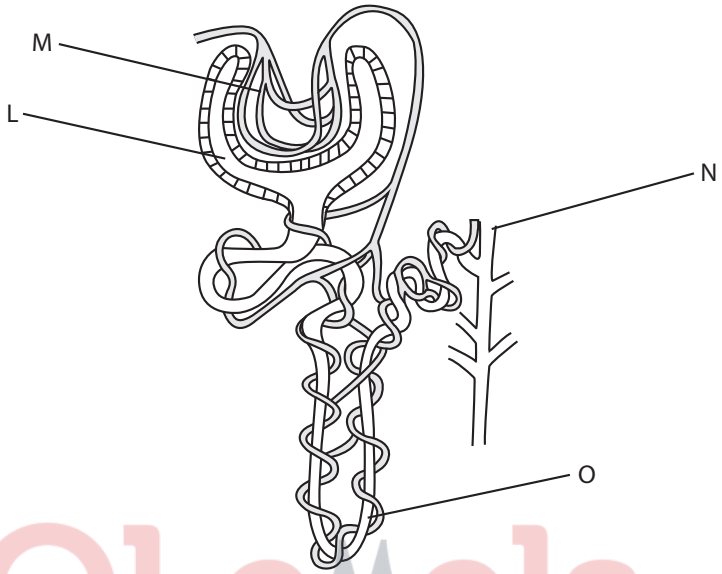
Type ID: HBI-NAME-RENAL-STRUCTURES

# 1. Question 6 · part a · 4 marks

November 2023 Paper 1

Worked-example retry · Attempt part a only.

**6** The diagram shows a kidney tubule.




(a) Name structures L, M, N, and O.

**(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L.

**(3)**

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## 1. Question 6 · part a · 4 marks · Working and answer

Working and answer

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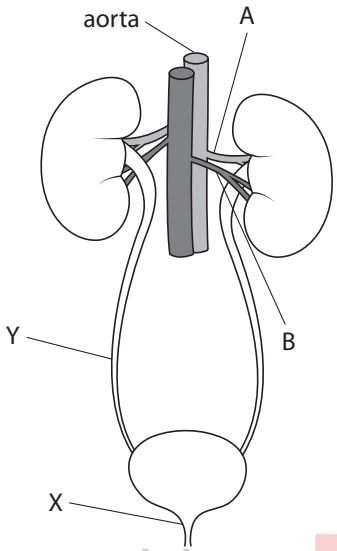
## 2. Question 8 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

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**8** The diagram shows part of the human renal system.




(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

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(iii) Compare the composition of the blood in vessels A and B. **(2)**

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## 2. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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# 3. Question 9 · part b · 4 marks

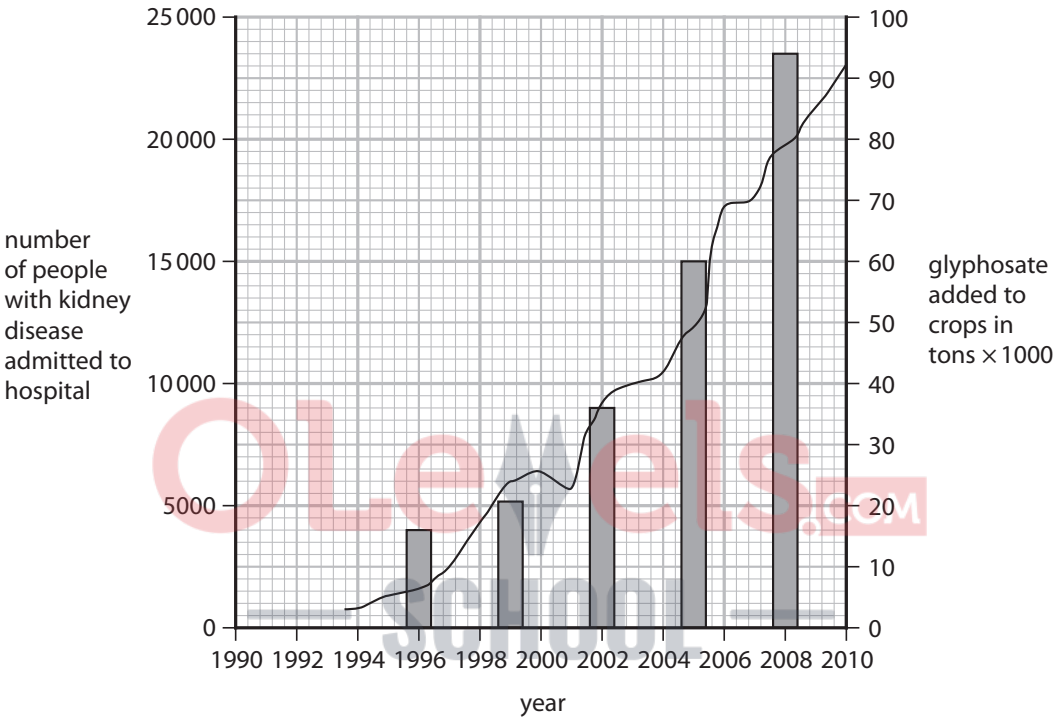
November 2020 Paper 1R

Further practice · Attempt part b only.

**9** (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.




(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

**(3)**

percentage increase = ..... %

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## 3. Question 9 · part b · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

**(1)**

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(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

**(4)**

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## 3. Question 9 · part b · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(c) There is evidence to show that glyphosate causes inflammation of the glomeruli and the kidney tubules.

Suggest how these symptoms will affect the normal functioning of the kidney and the waste product that it produces.

**(4)**

**(Total for Question 9 = 12 marks)**

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## 3. Question 9 · part b · 4 marks · Working and answer

Working and answer

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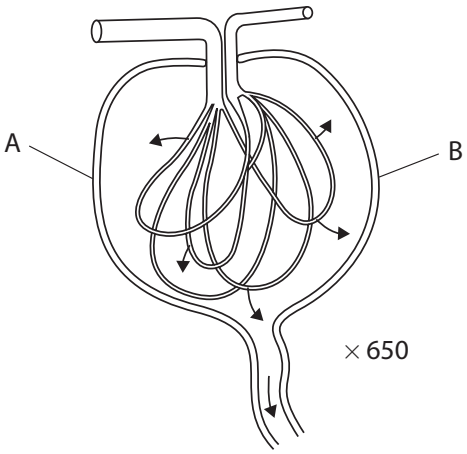
## 4. Question 1 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

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**1** (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).




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(i) Complete the diagram to show the missing parts of the kidney tubule.

**(3)**

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

**(3)**

width of Bowman's capsule = ..... mm

## 4. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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## 5. Question 7 · part a.i · 1 mark

June 2022 Paper 2R

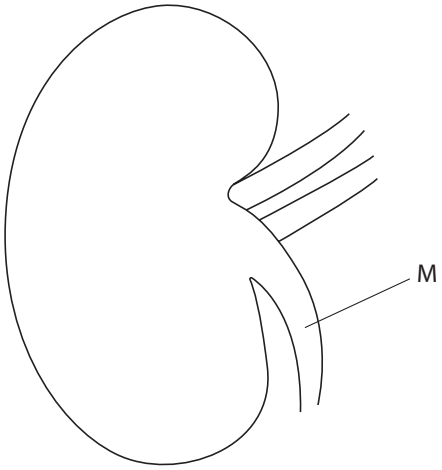
Further practice · Attempt part a.i only.

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**7** The diagram shows a kidney.




(a) (i) Which of these is the name of structure M?

**(1)**

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

**(1)**

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

## 5. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

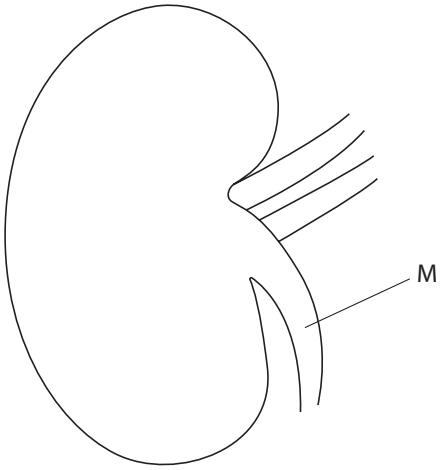
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# 6. Question 7 · part a.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

**7** The diagram shows a kidney.




(a) (i) Which of these is the name of structure M?

**(1)**

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

(ii) Which of these structures is connected to structure M?

**(1)**

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

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# 6. Question 7 · part a.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

**(3)**

1 .....

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2 .....

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3 .....

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(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

| Substance excreted during 24 hours | High protein diet Mass in g | Low protein diet Mass in g |
|---|---|---|
| total nitrogen compounds | 16.8 | 3.6 |
| urea | 14.7 | 2.2 |
| uric acid | 0.18 | 0.09 |
| ammonia | 0.49 | 0.04 |
| creatinine | 0.58 | 0.60 |

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

**(3)**

percentage difference = ..... %

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## 6. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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# 7. Question 7 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

**7** The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

**(1)**

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(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

**(3)**

| | Structure | | |
|---|---|---|---|
| **Function** | **glomerulus** | **proximal tubule** | **renal vein** |
| regulation of water level | | | |
| ultrafiltration | | | |
| reabsorption of glucose | | | |
| removal of blood from kidney | | | |
| supply of blood to kidney | | | |
| removal of urine from collecting duct | | | |

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# 7. Question 7 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

| Component | Percentage composition | |
|---|---|---|
| | first sample | second sample |
| ammonia | 0.05 | 0.08 |
| uric acid | 0.03 | 0.08 |
| ions | 1.34 | 1.46 |
| urea | 2.00 | 2.80 |
| water | 95.00 | 91.50 |

(i) Describe how urea is produced in the body.

**(2)**

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## 7. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

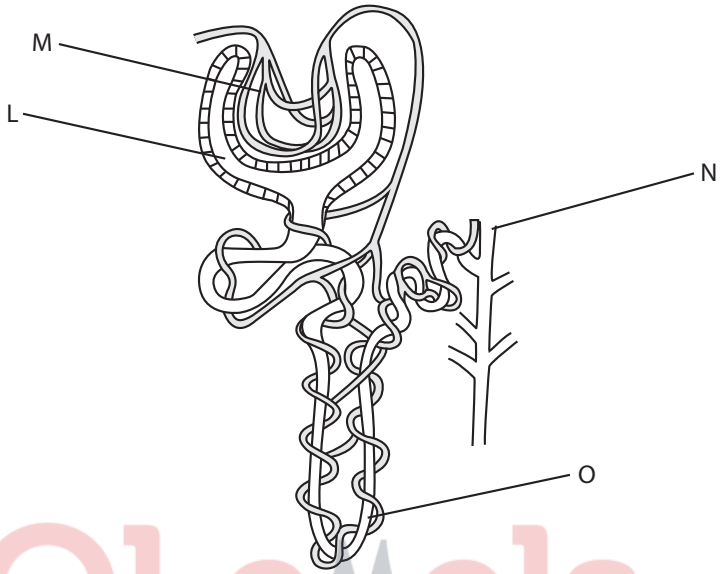
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# 8. Question 6 · part b.iii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iii only.

**6** The diagram shows a kidney tubule.




(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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## 8. Question 6 · part b.iii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Name **two** components of the blood that do not pass into L.

**(2)**

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

**(1)**

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

**(5)**

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**(Total for Question 6 = 15 marks)**

## 8. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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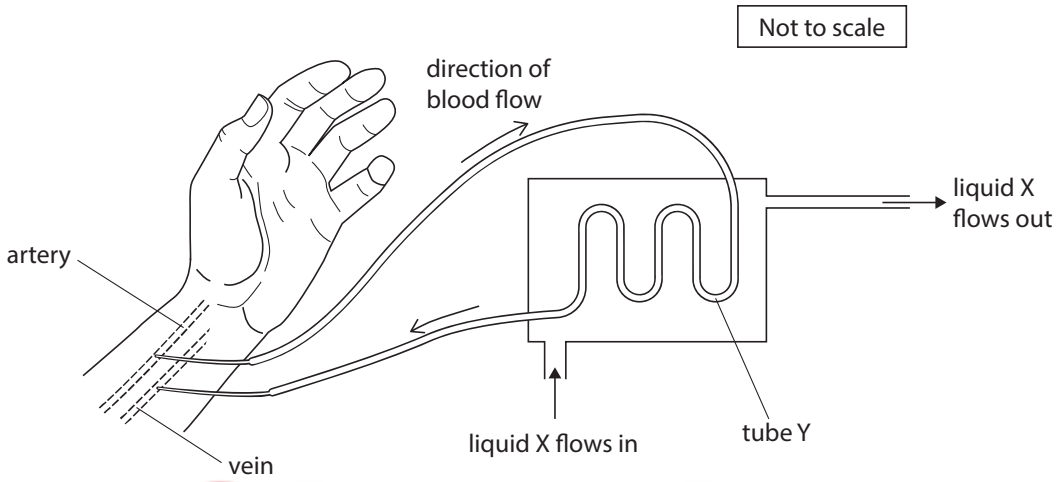
# 9. Question 5 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

**5** The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.




(a) (i) Explain why the blood flow is in the direction shown in the diagram.

**(4)**

(ii) State which part of the kidney nephron has the same role as tube Y.

**(1)**

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## 9. Question 5 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Explain why tube Y has a length of several metres in the dialysis machine.

**(2)**

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

**(1)**

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

**(3)**

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## 9. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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